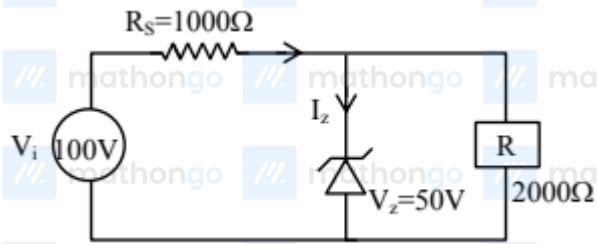


Q1 (20 July 2021 Shift 1)

For the circuit shown below, calculate the value of I_z :



(1) 25 mA

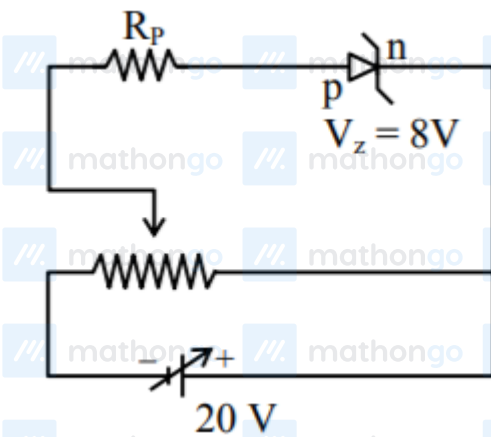
(2) 0.15 A

(3) 0.1 A

(4) 0.05 A

Q2 (20 July 2021 Shift 2)

A zener diode having zener voltage 8 V and power dissipation rating of 0.5 W is connected across a potential divider arranged with maximum potential drop across zener diode is as shown in the diagram. The value of protective resistance R_p is Ω .

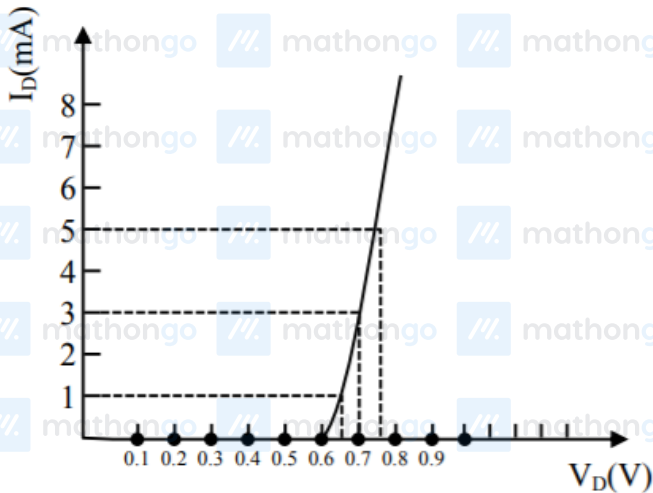


Q3 (20 July 2021 Shift 2)

For the forward biased diode characteristics shown

in the figure, the dynamic resistance at $I_D = 3 \text{ mA}$

will be _____ Ω



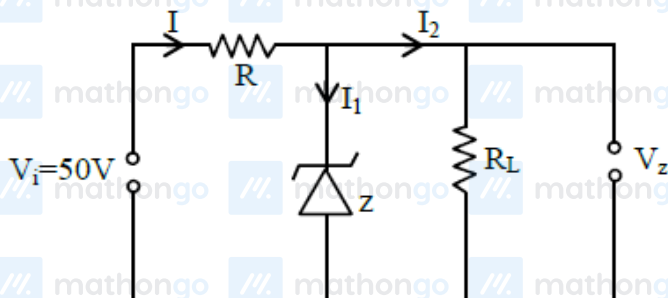
Q4 (22 July 2021 Shift 1)

Consider a situation in which reverse biased current of a particular P-N junction increases when it is exposed to a light of wavelength ≤ 621 nm. During this process, enhancement in carrier concentration takes place due to generation of hole-electron pairs. The value of band gap is nearly.

- (1) 2eV
- (2) 4eV
- (3) 1eV
- (4) 0.5eV

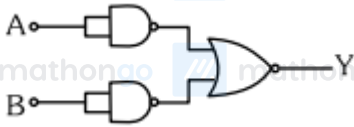
Q5 (22 July 2021 Shift 1)

In a given circuit diagram, a 5 V zener diode along with a series resistance is connected across a 50 V power supply. The minimum value of the resistance required, if the maximum zener current is 90 mA will be _____ Ω



Q6 (25 July 2021 Shift 1)

Identify the logic operation carried out.



- (1) OR
- (2) AND
- (3) NOR
- (4) NAND

Q7 (25 July 2021 Shift 2)

In a semiconductor, the number density of intrinsic charge carriers at 27°C is $1.5 \times 10^{16}/\text{m}^3$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22}/\text{m}^3$. The electron density in the doped semiconductor is $__ \times 10^9/\text{m}^3$

Q8 (27 July 2021 Shift 1)

A transistor is connected in common emitter circuit configuration, the collector supply voltage is 10 V and the voltage drop across a resistor of 1000Ω in the collector circuit is 0.6 V. If the current gain factor (β) is 24, then the base current is $__ \mu\text{A}$. (Round off to the Nearest Integer)

Q9 (27 July 2021 Shift 2)

Find the truth table for the function Y of A and B represented in the following figure.



- (1)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

(2)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

(3)

\$A\$	\$B\$	\$Y\$
0	0	0
0	1	0
1	0	0
1	1	1

(4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Answer Key

Q1 (1)

Q2 (192)

Q3 (25)

Q4 (1)

Q5 (500)

Q6 (2)

Q7 (5)

Q8 (25)

Q9 (2)

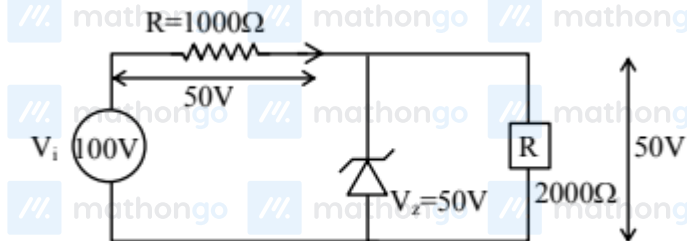
Hints and Solutions

MathonGo

Q1

$$I = \frac{50}{1000} = 50 \text{ mA}$$

$$R = 1000\Omega$$



$$I = \frac{50}{2000} = 25 \text{ mA}$$

$$I_Z = I_{1000} - I_{2000}$$

$$= 50 - 25 = 25 \text{ mA}$$

Q2

$$P = V_i$$

$$0.5 = 8i$$

$$i = \frac{1}{16} \text{ A}$$

$$E = 20 = 8 + iR_P$$

$$R_P = 12 \times 16 = 192\Omega$$

Q3

$$R_d = \frac{dV}{di} = \frac{1}{\frac{di}{dV}} = \frac{1}{\frac{5-1 \times 10^{-3}}{0.75-0.65}}$$

$$\frac{100}{4} = 25\Omega$$

Q4

$$\text{Band gap} = \frac{hc}{\lambda_0}$$

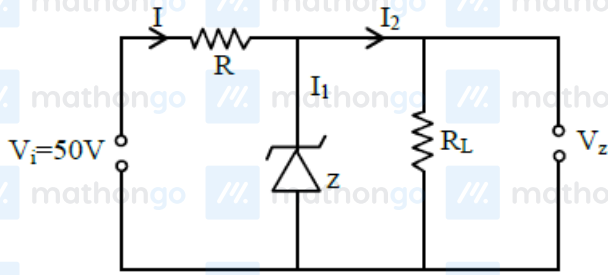
 λ_0 ; threshold wavelength

$$\text{Band gap} = \frac{1242 \text{ eV-nm}}{621 \text{ nm}} = 2 \text{ eV}$$

Q5

Hints and Solutions

MathonGo



Voltage across $R_L = 5 \text{ V}$

$$\Rightarrow i_2 = \frac{5}{R_L}$$

Also voltage across $R = 50 - 5 = 45 \text{ volt}$

$$\text{By } v = iR \Rightarrow R = \frac{v}{i} = \frac{45}{i_1 + i_2}$$

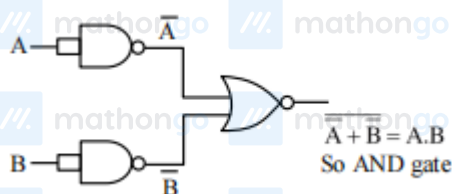
$$R = \frac{45}{90 \text{ mA} + \frac{5}{R_L}}$$

Current in zener diode is maximum when $R_L \rightarrow \infty$

($i_2 \rightarrow 0$ and $i_1 = i$)

$$\text{So } R = \frac{45}{90 \text{ mA}} = 500 \Omega$$

Q6



Q7

$$n_e n_h = n_i^2$$

$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16})^2}{4.5 \times 10^{22}}$$

$$= \frac{1.5 \times 1.5 \times 10^{32}}{4.5 \times 10^{22}}$$

$$5 \times 10^9 / \text{m}^3$$

Q8

Hints and Solutions

MathonGo

$$\beta = \frac{I_C}{I_B} = 24; \quad R_C = 1000$$

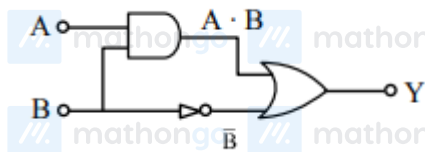
$$\Delta V = 0.6$$

$$I_C = \frac{0.6}{1000}$$

$$I_C = 6 \times 10^{-4}$$

$$I_B = \frac{I_C}{\beta} = \frac{6 \times 10^{-4}}{24} = 25 \mu A$$

Q9



$$Y = A \cdot B + \bar{B}$$

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1